DRONACHARYA
College of Engineering

KHENTAWAS, FARRUKHNAGAR, GURGAON, HR

DEPARTMENT OF CSE(AI&ML)

FIRST SESSIONAL EXAMINATION (JANUARY-JUNE-2026)

(GURUGRAM UNIVERSITY)

Course Code: PCC-AIML-403	Course Title: Design & Analysis of Algorithms	
Semester: IV	Maximum Marks: 70	Date: 09/03/2026
Branch: CSE(AI&ML)	Time: 03 Hours	Roll No.: 27240

Note: : *Q.1 is compulsory. All questions are compulsory. Each question carries equal marks (14 marks). Assume data, if any are missing.*

Q.1. ***Short Answer type questions based on REMEMBER/UNDERSTAND.***

(2 X 7=14 Marks)

a) Define algorithm? List any two essential characteristics of a good algorithm. **(CO1)**

b) Differentiate between time complexity and space complexity. What do you mean by best case, average case and worst-case time complexity of an algorithm? **(CO1)**

c) Differentiate between Big O, Omega, and Theta notations. **(CO1)**

d) What is Divide and Conquer approach? Explain with suitable example **(CO2)**

e) Analyze the time complexity of the following code fragment and determine the total number of executions of the statement k = k + 1 along with its Big-O complexity: **(CO4)**

```
for (i = 1; i <= n; i++)
{
   for (j = 1; j <= n; j++)
   {
      k = k + 1;
   }
}
```

f) The time complexity of factorial function for a number id Theta(n) ? **(CO4)**

g) How BFS is differ from DFS? **(CO5)**

SECTION-A

Q.2. Analyze the performance of an algorithm by explaining the key metrics used for evaluation, defining time and space complexity and their relationship, distinguishing between empirical and asymptotic analysis, and solve the recurrence relation $T(n)=T(n-2)+n$ for $n > 1$ given $T(0)=1$ and $T(1)=1$, deriving its time complexity. **(CO4)**

OR

Q.3. Solve the following recurrence relation using the back substitution method: $T(n) = k*T(n/k) + n^2$, when $T(1) = 1$ and k is any constant. **(CO4)**

4. Sort the sequence {22, 44, 33, 11, 77, 66, 55, 88, 100} using the Merge Sort algorithm by applying the Divide and Conquer strategy, showing all partitioning and merging steps, and state its best-case, average-case, and worst-case time complexities. **(CO3)**

OR

Q.5. Sort the sequence {22, 44, 33, 11, 77, 66, 55, 88, 100} using the Quick Sort algorithm, show the partitioning steps, and state its best-case, average-case, and worst-case time complexities along with how it follows the Divide and Conquer strategy. **(CO3)**

SECTION-B

Q.6. The classical matrix multiplication method requires $\Theta(n^3)$ time to multiply two $n \times n$ matrices. Explain how the Strassen's algorithm applies the Divide and Conquer strategy to reduce the number of multiplications. Multiply the given below two matrices (A & B) using Strassen's method. Write the recurrence relation for Strassen's algorithm and derive its time complexity. **(CO4)**

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 3 & 2 \\ 1 & 0 & 0 & 1 \\ 2 & 1 & 1 & 0 \end{bmatrix}$$

OR

Q.7. A university maintains the roll numbers of 60 students in sorted order in a one-dimensional array. Each roll number occupies 4 bytes of memory. The array is stored in contiguous memory locations starting from base address 2000.

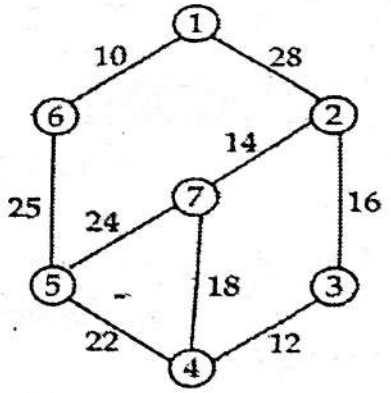
a) Calculate the total memory required to store the roll numbers.

b) Determine the memory address of the 25th student's roll number.

c) The Binary Search to search for roll number 48 in the following sorted array: [5,9,12,18,23,27,31,36,40,44,48,52,56,60,65,70]. Show all iterations (low, high, mid values).

d) Derive the recurrence relation for Binary Search and solve it to obtain its time complexity. **(CO4)**

Q.8. The Greedy Methods (Prim's and Kruskal's algorithms) are used to solve the Minimum Cost Spanning Tree problem. Explain the concept of minimum spanning trees. Solve the following graph using Prim's algorithm. Also analyse its complexity. **(CO3)**



OR

Q.9. Considering a collection of cities A, B, C, and D with the following directed distances between them: **City separations:**

- A to B = 11, A to C = 13, A to D = 19
- B to A = 5, B to C = 10, B to D = 9
- C to A = 6, C to B = 12, C to D = 11
- D to A = 8, D to B = 8, D to C = 10

Treat the cities as vertices of a weighted directed graph. Using **Dijkstra's Algorithm** for single source shortest paths, and show how does it use the greedy method to determine the shortest paths from city **A** to all other cities.

a) Represent the graph using a suitable diagram or adjacency matrix.

b) Show all intermediate steps of Dijkstra's algorithm (distance table updates).

c) Construct the shortest path tree. **(CO3)**